

Genetic master's document

Immortality

Everything that is extremely important to overcome the weaknesses and live as much as possible

Sérgio F · 03-10-2026

Synthesis of all genetic reports (profile with 106 sections, 86 thematic reports, 8 external reports, pharmacogenomics of 632 drugs, 999 supplements, 812 foods), plus 5 new analyzes to the raw DNA and imputation. Only genes: does not use blood tests or exams. Each point has the strength of evidence (strong, moderate or weak) and the confidence of the data (chip, imputed or PGS). It is neither a diagnosis nor a prescription. All actions are marked as suggestion (not applied).

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1 page summary: the 10 actions that most change the outcome

Document-master of action, made from all genetic reports (8 external reports, 86 thematic reports, the complete profile with 106 sections, the genetic plan, the pharmacogenomics of 632 drugs, 999 supplements and 812 foods) and new analyzes to the raw DNA and the imputation. Only genes. Order : probable impact on mortality and aging × strength of evidence × possibility of acting. All actions are suggestion (not applied), to decide with the doctor.

#	Action	Because (genes)	Evidence
1	ApoB and LDL as low as possible, since now and for all life; zero tobacco, vaping or smoking	9p21 (CDKN2A/B) risk homozygous in the 5 SNPs of the haplotype (4 read on the chip and 1 imputed with DR2 1; r 0.92-0.998 connection confirmed again); CETP of high activity; Low polygenic HDL	Strong (9p21 is the strongest coronal common locus)
2	Protecting the liver: zero alcohol, no smoking, low hepatic fat; confirm Pi*MZ; base and periodic elastography; hepatitis B vaccine	SERPINA1 Pi*Z heterozygous (chip, single P/LP plausible in the read genome) + PNPLA3 I148M + MBOAT7; HSD17B13 protector attenuation	Strong (large effect variables, direct chip)
3	Strength training 3x/week with progression, protein ~140 g/day, creatine; DEXA to follow lean mass	P10 appendicular lean mass, p84 fractures, p75 fragility, p14 genetic physical activity (PGS)	Moderate, very actionable
4	Mood, attention and sleep accompanied closely; TCC for insomnia; melatonin 0.3-1 mg; monitoring mood with isotretinoin	Major depression p95, ADHD p94 and p97, insomnia (129 loci) p89, schizophrenia p88, bipolar p79 (PGS)	Moderate
5	Colonoscopy of quality at 45 years maximum, more FIT (hidden blood immunochemistry research) from 40; fiber: more is always better	Colorectal cancer: 95 variant p94,3, p78,5, p82,3 early onset, p92,7 (PGS; replaces §24 reading)	Moderate
6	Preventing thrombosis in surgery, immobilization and long journeys; knowing the signs of alarm	Venous thrombosis p86 (297 variants) and loci p95,1; pulmonary embolism p90,8 F11 rs2289252 T/T (chip)	Moderate
7	Pharmacogenetic card in the clinical process and in the cell phone	CYP2C9 *1/*3 + VKORC1 T/T (varfarin, NSAIDs), NAT2 slow, UGT1A1 *28, CYP3A5 expressor (rapamycin), CYP2D6 without copy number, HLA-B*58:01 per tipar	Strong (CPIC/DPWG)
8	Complete genome to clinical grade 30 (secondary findings ACMG, pharmacogenetics with CYP2D6, HLA); clonal hematopoiesis panel later	The chip reads 6.9% of known P/LP variants in the genome (21% in ACMG genes); does not see GSTM1/GSTT1, BMPR2, Lp(a) KIV-2 or acquired clones	Strong (proven technique)
9	Vaccines in day: pneumococcal conjugated (already, by rapamycin), annual flu, COVID-19, recombinant zone, hepatitis B	Pneumonia p84,2, urinary infection p93,8, otitis p85,3 (PGS); FCGR2A H131R heterozygous	Moderate
10	Base exams once: echocardiogram (aortic valve and left ventricle thickness), OCT of the optic nerve with intraocular pressure, spirometry, audiogram	Aortic valve: p89-95 aortic stenosis (492 loci p95,1; other p88,8 score); hypertrophic cardiomyopathy p76-89 (PGS); glaucoma p97,9; emphysema p96,2 (with Pi*MZ); tinnitus (loci) p99,8	Moderate

In a sentence: good foundations (APOE ε3/ε3, FOXO3 of longevity in homozygosity, polygenic coronary risk p5, low genetic glucose, almost all cancers with low risk, PGS of longevity p92) with four focal weaknesses that are circumvented with prevention: arteries (9p21), liver (Pi*MZ + PNPLA3), brain and mood, and muscle. None in the read genome indicates a severe monogenic disease.

Almost everything is primary prevention with decades of margin.

Polygenic scores update (03-10-2026). The scores were recalculated with more published recent scores (the same method and the same reference population). This box prevails over the previous values cited in the text.

- Coronary disease: new scores p54, **p18** and p5, 5; with the previous ones, the consensus is p18. It remains low, less extreme than p5.
- Venous thrombosis: 3 new scores give p24 and p71 (genomic) and p80 (10 SNPs); with the previous ones (p86, p84 and p54), the consensus is **p76**. It goes from high to above average; the practical rules remain.
- Bladder cancer: new genomics give p57 and p27 and 24 SNPs p86; with the previous (p90, 15 SNPs), the consensus is **p72**. It goes from high to intermediate-high; Slow NAT2, zero tobacco, avoid aromatic amines and pioglitazone are not recommended.
- Lean mass: new p32 (genomic) and p3,5 scores; with the previous ones (**p27**, p30 and p10), the consensus is p27. Under the average, lower than the p10 cited; strength and protein training remain.
- Fractures: the only new score (45 SNPs) gives p16, against p84 in the previous one; the genetic risk of fracture becomes uncertain and bone densitometry decides.
- Alzheimer's disease: the new genomics give p75 and p85 (before p52); with APOE ε3/ε3, reading goes from medium to medium-high (genomic consensus p75). The suggestions remain.
- Confirmed: high colorectal (p80), prostate (p5,5), melanoma (p18) and low atrial fibrillation (p9), depression (p85), glaucoma (p85) and high insomnia (p80), typical 2 diabetes (p58).

1. Main causes of death and disability predicted by genes

How it was ordered. The starting point is the real frequency of each cause of death and disability in adults of the reference population. In order: cardiovascular disease, cancer, neurological disease and dementia, respiratory disease and infections, liver disease, external causes and mental health. To this base joins the genetic deviation of the holder. The result is not a prediction of when or what: it is an order from which prevention yields more.

All prevention, screening and avoidance lines of this section are suggestion (not applied).

Subtitle. Evidence: strong, moderate or weak. Confidence of the data: chip (genotype read directly), imputed (estimated from the chip, with quality $DR2 \geq 0.8$) or **PGS** (polygenic score with percentile compared to the reference population; describes a trend, not a diagnosis).

1.1 Atherosclerotic cardiovascular disease (infarction, stroke of large vessels, aneurysm, peripheral arterial disease)

Genetic risks	9p21 (CDKN2A/B) risk homozygous: rs1333049 C/C, rs10757278 G/G, rs4977574 G/G, more rs1333042 G/G (chip) and rs1556516 C/C (imputed, $DR2 \geq 1$) in the same haplotype (linking $r \geq 0.92-0.998$ in 633 reference genomes). High activity CETP (TaqIB B1B1); HDL genetic p17-20. <i>Strong evidence for 9p21, moderated for the rest.</i>
Attenuants	Low polygenic coronary: metaGRS p5, GPS p30 (PGS). High genetic Lp(a) unlikely: rs10455872 A/A and rs3798220 T/T (chip), although PGS is in p75. Stroke in p19, stroke lacunar p14, aneurysm of the abdominal aorta p26-42, LDL polygenic p20, 5. Group O probable (ABO rs505922 T/T): less arterial and venous thrombosis.
Interpretation	Both are true: the global polygenic risk is low, but there is a strong focal point in the arterial wall. 9p21 does not act through cholesterol, so the defense is less atherogenic particles for longer.
Prevention	ApoB and LDL as low as possible, without lower limit (maintaining current lipid therapy; resuming the inclisiran or another anti-PCSK9 is a point to discuss). Zero tobacco, vaping and smoked cannabis. Fiber (more is always better), Omega-3 to 3.2-4 g/day, Mediterranean pattern, aerobic and strength exercise, regular sleep.
Tracking	Coronary calcium score between 35 and 40 years; if it is 0, repeat 5 in 5 years; if it is > 0 , coronary angio-TC and intensify. Ankle-arm index to be discussed from 50.
Alert signs	Pain or tightening of the chest in effort or at rest, lack of disproportionate air, pain in the legs when walking that relieves upon stopping, weakness or sudden dormancy on one side, sudden alteration of speech or vision (binding to the emergency number).
Avoid	Any form of smoking; periods without lipid therapy; ketogenic diets rich in saturated fat; NSAIDs of chronic use (same as CYP2C9*3).

1.2 Cancer (colorectal first; then bladder, pancreas and esophagus)

Genetic risks	Colorectal: 95 score variants p94,3,p78,5,p82,3 early onset,p92,7 (PGS; moderate evidence). P90 boxing with slow acetylator NAT2 (rs1801280 C/C, rs1799930 G/G, rs1799931 G/G; chip): slow NAT2 only counts in the presence of aromatic amines (smoker, paints, rubber, solvents). Pancreas p83, esophagus p85, glioma p91 (PGS; weak to moderate evidence).
Attenuants	Genome prostate p0,9 without the strong loci of 8q24; melanoma p6,5 and p25 with clean MC1R; cell squamous p3,5; kidney p0,8; non-Hodgkin lymphoma p1,1. No P/LP variants of hereditary cancer genes read (BRCA1/2, Lynch, APC, MUTYH) within that the chip covers. CASP8 -652 6N of heterozygous (new; "protective" classification, weak evidence). Y loss in mosaic: favourable genetics (p86).
Prevention	High and varied fiber; little processed meat and very low red meat; zero alcohol (already fulfills); zero tobacco; keeping the visceral fat mass low; vitamin D adequate; regular exercise. In the profession and in the hobbies, avoid aromatic amines and solvents (slow NAT2).
Tracking	FIT annual (or 2 in 2 years) from 40; quality colonoscopy at 45, with the range defined by the findings (typically 5-10 years). Urine for microscopic hematuria regularly from 40 (NAT2 + bladder p90; weak evidence, low cost). Skin: annual observation by dermatology while doing isotretinoin, and after 2 in 2 years or earlier if there are signs. PSA by shared decision from 50 (prostate genetic risk is low).
Alert signs	Blood in the feces or black faeces, persistent change in intestinal transit, unexplained anemia, blood in urine, loss of mass without explanation, painless jaundice, difficulty swallowing, signal on the skin that changes.
Avoid	Smoke; processed meat; professional exposure to aromatic amines and solvents; folic acid in any supplement (prefer 5-MTHF); telomerase activators; GH/IGF-1 secretagogues.

1.3 Chronic liver disease (fibrosis, cirrhosis, hepatocellular carcinoma)

Genetic risks

SERPINA1 Pi*Z heterozygous (rs28929474 C/T; chip; confirmed de novo; is the only plausible P/LP in the ClinVar screening of the entire genome read). **PNPLA3 I148M** C/G, **MBOAT7** C/T (chip). PGS of ALT p87,0 and GGT p94,0. *Strong evidence for Pi*Z and PNPLA3 as fibrosis modifiers.*

Attenuants	HSD17B13 rs72613567 T/TA heterozygous (imputed, DR2 1; confirmed in imputation): reduces the progression to steatohepatitis and fibrosis. TM6SF2 C/C (without risk). HFE C282Y G/G (without hemochromatosis). Typical polygenic liver fat (p65–69).
Interpretation	Pi*MZ alone increases little risk; added to PNPLA3 and any aggressor (alcohol, visceral fat, drugs), clearly increases. The point is the liver as an organ to protect for decades.
Prevention	Zero alcohol; no smoking; low visceral fat; filtered coffee is acceptable; ferritin below 100; no iron supplements; paracetamol at the minimum effective dose (maximum 2 g/day, as in the genetic plan); hepatic surveillance with isotretinoin, rapamycin and other drugs; hepatitis B vaccine if not immune.
Tracking	Confirm the Pi*MZ (alfa-1-antitripsina serum + Pi phenotype) already. Transient base liver elastography, after 2 in 2 or 3 in 3 years and whenever it changes some risk factor. If advanced fibrosis appears, 6 ultrasound in 6 months.
Alert signs	Persistent tiredness, itching, dark urine, yellow eyes, swelling of the belly or legs, easy black stains.
Avoid	Alcohol; herbal supplements with liver risk; high doses of vitamin A or retinol; iron; hepatotoxic drugs without surveillance (slow NAT2 is the worst case).

1.4 Venous thromboembolism (deep venous thrombosis, pulmonary embolism)

Genetic risks	Venous thrombosis p86 (297 variants) and p84,4; loci of VTE p95,1; pulmonary embolism loci p90,8 F11 rs2289252 T/T (chip). <i>Moderate evidence.</i>
Attenuants	Factor V Leiden and negative prothrombin G20210A (chip). Group O probable.
Prevention	To move 1 in 1 hour on long flights and trips; hydration; compression stockings on flights above 4 hours; always inform the genetic risk before surgeries, immobilizations and gypsum (prophylaxis with heparin to be discussed); any estrogen or testosterone exogenous only with evaluation of thrombotic risk.
Tracking	There is no screening for exams; prevention is done in risk situations.
Alert signs	Swelling, warm and painful legs on one side; sudden breathlessness; chest pain upon breathing; coughing with blood (urgency).
Avoid	Prolonged immobilization; dehydration; testosterone or other androgens without evaluation; tobacco.

1.5 Brain: small vessels, dementia and cognitive decline

Genetic risks	White matter p92 lesions (355 variants; PGS, moderate evidence); genetic homocysteine p92 (MTHFR C677T G/A + A1298C T/G; chip).
Attenuants	APOE ε3/ε3 (rs429358 T/T imputed DR2 0.99 + rs7412 C/C chip). Genomic Alzheimer p52 (values the genomic; see contradictions). Parkinson p7,7; MAPT H2/H2 Stroke lacunar p14; intracerebral hemorrhage p18.
Prevention	Everything that protects the arteries (section 1.1); sleep of 7–9 hours and regular (the clean glymphatic system during deep sleep); aerobic exercise; hearing protection (hearing loss is the major modifiable risk factor for dementia and genetic tinnitus is high); active folate, B12 and B2 (MTHFR); continuous learning and social life.
Tracking	Cognitive assessment between 45 and 50; Cerebral RM only if there is clinical indication.
Alert signs	Oblivion that interferes with daily life, disorientation, personality changes; stroke signals (1.1 section).
Avoid	Chronic sleep deprivation; noise without protection; anticholinergics of chronic use; cranial trauma (capacet).

1.6 Mental health, attention and external causes

Genetic risks	Major p95 Depression; ADHD p94 and p97; insomnia p89; schizophrenia p88; bipolar p79 (PGS; moderate evidence; psychiatric PGS explain a small part of the risk). Duration of genetic sleep p17. Beverages per week p94, 6, but does not drink alcohol.
Interpretation	It is the area with the highest percentiles. ADHD is associated with more accidents and injuries, and depression is treatable and predictable. It is not a destination: it is a reason to have a network and a plan.
Prevention	Regular follow-up with a doctor or psychologist; exercise (is antidepressant); morning light; regular sleep; TCC for insomnia before any hypnotic; monitoring the mood during isotretinoin (the PGS of depression add to the warning of the bull); defensive driving and without cell phone.
Tracking	Brief (PHQ-9, GAD-7) 6 questionnaires in 6 months or whenever there is a change; formal evaluation of ADHD was never performed.
Alert signs	Sadness or lack of persistent pleasure for more than 2 weeks, ideas of death (seeking help in the day itself: support line or emergency number), persistent insomnia, sleepless euphoria.

Avoid	Alcohol and cannabis; unsupervised stimulants; high dose melatonin (deregulates the circadian rhythm; use 0.3–1 mg).
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1.7 Respiratory infections and sepsis (especially in advanced age)

Genetic risks	Pneumonia p84,2, otitis p85,3, urinary infection p93,8 (PGS); FCGR2A H131R heterozygous (chip). Rapamycin and advanced age add up.
Attenuants	COVID-19 severe p4,7; sepsis p23; MBL2 A/O (intermediate, without deficiency), normal TLR2 and TLR4; Low genetic CRP (p12).
Prevention and screening	Conjugated pneumococcal (PCV20 or PCV21) already, due to intermittent immunosuppression; annual flu; COVID-19 according to the recommendations in force; RSV from 60 (or earlier indicated); earlier recombinant zone (rapamycin); oral hygiene; no smoking.
Alert signs	Fever with lack of air, confusion or prostration; urinary burning with fever or low back pain.
Avoid	Rapamycin during acute infections (suspend and talk to the doctor); contact with unprotected patients during rapamycin periods.

1.8 Fragility, falls and fractures (disability)

Genetic risks	P10 appendicular lean mass; p84 fractures; p75 fragility; p14,2 genetic physical activity (PGS; moderate evidence).
Attenuants	ACTN3 R577X T/C (fast fibers present); aerobic resistance p92; grip strength p52,4 (typical); genetic resting heart rate p4,4 (favourable).
Prevention	3 times a week with progression; distributed protein (~140 g/day, 30–40 g per meal) with sufficient leucine; creatine 5 g/day (already taken); impact and jumps to the bone; balance from 50; vitamin D and K2; protect lean mass during retatrutide.
Tracking	DEXA of whole body (lean mass + bone density) already, as a base, and after 2 in 2 years or when changing the medication; grip strength and annual functional tests from the 50.
Alert signs	Rapid loss of strength, falls, bone pain after small trauma.
Avoid	Prolonged protein deficit; rapid mass loss without strength training; sedentary lifestyle.

1.9 Aortic valve and ascending aorta

Genetic risks	Aortic valve: p89–95 aortic stenosis (492 loci p95,1; another p88,8 score; PGS, moderate evidence); the strong alleles of Lp(a) are absent. Hypertrophic cardiomyopathy above average: p76 (374 thousand variants) and p89 (27 variants), FHOD3 rs2644262 T/C (PGS; weak to moderate evidence). An ascending aorta: genetic diameter p79 (PGS; genetic finding, without proper examination).
Prevention	ApoB low; no smoking; avoid quinolones when there is an alternative (associated with aneurysm and dissection).
Tracking	Base echocardiogram already, by the aortic valve and by the genetics of hypertrophic cardiomyopathy (left ventricular wall thickness); repeat to 40, and after 5 in 5 years (or before, if breath arises).
Alert signs	Lack of effort, exhaustion, chest pain; sudden lancinating pain in the breast or back (urgency).
Avoid	Fluoroquinolones without need; high dose stimulants.

1.10 2 type diabetes and metabolism

Genetic risks	2 p57 type diabetes (after imputation); TCF7L2 rs7903146 C/T (chip); trunk fat p98,4 (PGS).
Attenuants	HbA1c genetic p6, 6 and p8, 4; typical genetic triglycerides (p68, 2).
Prevention	Keeping the visceral fat low (the trunk fat score is the weak point); strength + aerobic; fiber; the current stack (imeglymine, acarbose, dapagliflozin, retatrutide) already covers this pathway.
Tracking	Annual metabolic evaluation (not part of this genetic document).
Avoid	Liquid sugars; sedentary lifestyle; lean mass loss.

1.11 Lung (COPD and emphysema)

Genetic risks	P96,2 (PGS) emphysema with Pi*MZ COPD p43,6; having smoked p2 (low-risk genetics to start the habit).
Interpretation	Without smoking the risk is small; with smoking, Pi*MZ accelerates lung destruction.

Prevention and screening	Zero smoke, vaping and passive smoke; FFP2 mask with dust and fumes; base spirometry already, after 40 and 5 in 5 years.
Alert signs	Lack of progressive air, chronic cough, wheezing.

1.12 Eyes, hearing and mouth (sensory disability)

Genetic risks	Glaucoma p97, 9 (PGS); tinnitus 89.7, 94.6 and 99.8 (PGS and loci); gingivitis p89, 7. CFH Y402H C/T (chip; macular degeneration, modest risk).
Prevention	Hearing protection (concerts, tools, hearing aids below 60% of the volume); sun glasses; lutein and zeaxanthin; rigorous oral hygiene.
Tracking	Ophthalmological examination with intraocular pressure and OCT of the optic nerve already, after 2 in 2 years until 40 and annual from 40. Base audiogram already, after 5 in 5 years and annual from 50. Dentist and hygiene of 6 in 6 months.
Alert signs	Peripheral visual field loss, red eye and painful with halos (urgency); sudden hearing loss (urgency in 72 hours, especially with tadalafil); bleeding gums.

2. 84 map of aging mechanisms with action priority

The 84 mechanisms of the target list, with the genetic state of the previous synthesis and a new action priority. High = weak point with impact action. Mean = mixed point with useful action. Maintain : genetic strength. Monitoring : typical. — = without genetic reading or outside the scope.

Counting: high 14, medium 17, 15 to maintain, 9 to watch, 29 without reading or outside the scope. The 47 mechanism of the list appears as a "mitochondrial redox cofactor" and is outside the scope.

#	Mechanism	State	Priority	Main action (suggestion, not applied)	Genetic findings
Main brands					
1	Genomic instability	Favourable	Keep	Strength to maintain	Common routes of repair without alleles of lower efficiency 0 variants P/LP plausible in 120 genes of hereditary disease; low predicted Y loss (dose in the 86th percentile). TP53 Arg72/Arg72 is a compromise.
2	DNA Damage and DNA Repair Deficit	Mixed	High	Avoid aromatic amines and smoke (NAT2, GSTM1); radiation only when useful	XRCC1, XRCC3, ERCC2 and XPC with common genotypes; but NAT2 slow acetylator, probable zero GSTM1 and GSTA1*B/*B pass through more fertilizers (aromatic amines, aldehydes); ATM D1853N A/A (radiation).
3	Hayflick's Telomeres and Limit	Unfavourable	Medium	Lifestyle; nothing of telomerase activators	Genetically shorter telomeres (87 loci in the 25th percentile; 7 SNPs in 24; TERT rs10069690 T/T. TERT rs2736100 C/C elongates, but with more glioma and clonal expansions (commitment).
4	Epigenetic Changes	Typical	Monitor	No specific action	Intrinsic clock (IEAA) in the 48th percentile, GrimAge in the 7 (indicative), PhenoAge in the 78 and Hannum in the 68; MTHFR compound heterozygous and genetic homocysteine in the 92nd percentile (methylation).
5	Proteostasis loss	Mixed	High	Protecting the liver (Pi*Z); sleep and exercise	Favourable neuron (APOE ε3/ε3, MAPT H2/H2, without GBA1 or LRRK2 G2019S); unfavourable liver (alfa-1 antitrypsin Z protein folds poorly and accumulates).
6	Autophagy / Deregulated Macroautophagy	Favourable	Keep	Strength to maintain	FOXO3 homozygous for the 4 alleles of longevity (active to autophagy); ATG16L1 T300A and MTOR rs2295080 heterozygotes (on average).
7	Nutrient Sensing Deregulation	Favourable	Keep	Strength to maintain	FOXO3 homozygous longevity; IGF-1 genetic in the 7th percentile; GHR d3 with 1 copy (rs6873545 T/C); IGF1R rs2229765 G/A. BCAT2 C/C (BCAA a little higher, small effect).
8	Mitochondrial dysfunction	Favourable	Keep	Strength to maintain	PGC-1α Gly482Gly, normal NDUFB6, β-oxidation without pathogenic variants, mtDNA HV without classical variants. Small: SIRT3 V208I and UCP1 –3826 heterozygotes.
9	Cell Senescence	Unfavourable	High	ApoB/LDL as low as possible; zero smoke (9p21)	9p21 (CDKN2A/B) homozygous for the risk haplotype in 5 SNPs (4 on chip, 1 imputed); CDKN1A (p21) G/A; GDF-15 genetic in the 83rd percentile; IL6 –174 G/G (plus SASP). TP53 Arg72 favors apoptosis instead of senescence.
10	Exhaustion of Staminary Cells	Mixed	Medium	CHIP panel to 40–50	Telomeric reserve slightly below the average (the 25th percentile), but without JAK2 46/1, SH2B3 C/C and CHIP score in the 30th percentile.
11	Intercellular Communication	Typical	Monitor	No specific action	Favourable hormones (genetic testosterone in the 89th percentile, low IGF-1) with high

#	Mechanism	State	Priority	Main action (suggestion, not applied)	Genetic findings
					SHBG (the 87th percentile); typical cytokines.
12	Chronic Inflammation	Mixed	Medium	Low visceral fat, omega-3, sleep	Low genetic CRP (the 12th percentile), IL6R Asp358Ala (1 copy), TNF –308 G/G and HLA autoimmune clean; against GlycA in the 90th percentile, IL6 –174 G/G and gingivitis in 90.
13	Dysbiosis	Favourable	Keep	Strength to maintain	FUT2 secretor, persistent lactase, inflammatory bowel disease in the 2–9 percentiles, celiac in the 4, NOD2 without variants; CARD9 S12N T/T (antifungal immunity) and normal FMO3 (TMAO).
Epigenetics, nucleus and genome					
14	Epigenetic Derivative	Typical	Monitor	No specific action	No known causal gene; the loci of epigenetic clocks give a typical intrinsic clock (the 48th percentile).
15	Loss of Epigenetic Fidelity	No reading	—	No useful genetic reading	No germinal variants with known effect are measured with epigenetic clocks in the blood, not with the chip.
16	Loss of Heterochromatin	No reading	—	No useful genetic reading	Without direct reading; SIRT6 of the centenaries (centSIRT6) is not read with confidence.
17	Epigenetic Memory Deregulation	No reading	—	No useful genetic reading	No genetic reading.
18	Nucleosome and Histone alterations	No reading	—	No useful genetic reading	No useful genetic reading (rare variants of histones and their modifiers are not on the chip).
19	Aberrant Transcriptional Reprogramming	No reading	—	No useful genetic reading	No genetic reading.
20	Disorganization of Nuclear Architecture	No reading	—	No useful genetic reading	Without genetic reading; LMNA has no plausible P/LP variants (the only reading was probe artifact).
21	Deterioration of the Nuclear Blade and the Domains Associated with the Blade	No reading	—	No useful genetic reading	Without genetic reading (LMNA without P/LP; centSIRT6, which reinforces the binding to lamina A, is not read).
22	Retrotransposing activation	No reading	—	No useful genetic reading	Without genetic reading; centSIRT6 (which stops LINE-1) is not read.
23	Endogenous Retrovirus dysregulation	No reading	—	No useful genetic reading	The chip does not read the endogenous retrovirus insertions.
24	Somatic Mutation Accumulation	Mixed	High	FIT to 40, colonoscopy to 45 at most; fiber	Common favourable repair and protective pigmentation (in childhood in the 1st percentile, low skin cancers); but weak II phase detoxification, TERT axis (glioma in the 91st percentile) and colorectal cancer above the median in the complete scores (94.3 and 78.5; polyps in the 92.7 loci).
25	Somatic / Tecdual Mosaicism	Favourable	Keep	Strength to maintain	Y loss in the 37th percentile and Y dosage in the 86 (less predicted loss).
26	Clonal hematopoiesis (CHIP)	Typical	Monitor	CHIP panel in blood at 40–50	Typical to low predisposition: without JAK2 46/1, SH2B3 C/C, CHIP score in the 30th percentile; TCL1A G/G (without the protective allele) and TERT rs7705526 heterozygous. The real state (acquired clones) is not seen in the chip.
27	Dysfunction of Primary Cyls	No reading	—	No useful genetic reading	No genetic reading.
RNA and proteostasis					

#	Mechanism	State	Priority	Main action (suggestion, not applied)	Genetic findings
28	RNA dysregulation	No reading	—	No useful genetic reading	No genetic reading.
29	Critical Transcription	No reading	—	No useful genetic reading	No genetic reading.
30	Splicing Errors	No reading	—	No useful genetic reading	Without genetic reading (the rare splicing variants are almost not on the chip).
31	Increased Transcriptional Elongation Velocity	No reading	—	No useful genetic reading	No genetic reading.
32	Transcriptional Decline Depending on Gene Size	No reading	—	No useful genetic reading	No genetic reading.
33	Protein Aggregation	Mixed	Medium	Sleep and exercise	Favourable brain: APOE ε3/ε3, MAPT H2/H2, TTR V122I absent, Parkinson in the 8th percentile; Alzheimer's without APOE of 39 variants in 91 (genomic no 52). Liver: protein Z.
34	Molecular Platerone Failure	No reading	—	No useful genetic reading	No direct genetic reading.
35	Proteasome dysfunction	No reading	—	No useful genetic reading	No genetic reading.
36	Lysosome dysfunction	Mixed	Medium	No specific action beyond lifestyle	GBA1 and LRRK2 G2019S absent and Parkinson low; but LRRK2 rs76904798 T/T, TMEM175 and GPNMB with risk alleles.
37	Autophagy Mediated by Reduced Chaperones	No reading	—	No useful genetic reading	No direct genetic reading (LRRK2 G2019S absent).
38	ER-fagia Defective	No reading	—	No useful genetic reading	Without genetic reading, it is one of the pathways that eliminate Z protein in the liver.
39	Remodelling of the Endoplasmic Reticulum	No reading	—	No useful genetic reading	No genetic reading.
40	Endoplasmic Reticulum Disorder	Unfavourable	High	Confirm Pi*MZ; elastography; zero alcohol	The Z protein (PiMZ) accumulates in the reticulum of the hepatocyte; PERKrs7571971C/C (favourable) does not compensate.
Mitochondria, redox and peroxisomes					
41	Oxidative Phosphorylation Decline	Favourable	Keep	Strength to maintain	PGC-1α Gly482Gly, NDUFB6 and COQ2 without risk variants, mtDNA without classical variants.
42	Changes in Mitochondrial Dynamics	No reading	—	No useful genetic reading	No useful genetic reading.
43	Mitochondrial Defective Quality Control	Favourable	Keep	Strength to maintain	FOXO3 homozygous longevity (mythophagy); no variants read at risk in the pathways of mitophagy.
44	Mytochondrial Calcium Deregulation	No reading	—	No useful genetic reading	No genetic reading.
45	Mutations in the DNA Mitochondrial	Favourable	Keep	Strength to maintain	Inherited: haplogroup HV without m.3243A>G, m.8344A>G, m.11778G>A or m.14484T>C. Somatic mutations and heteroplasma are not seen.
46	Release of Mytochondrial DAMPs	Typical	Monitor	No specific action	HMGB1 (loci) in the 47th percentile; without direct reading.
47	47 mechanism of the list	Out of scope	—	Out of the scope of this document	Out of the scope of these reports, by rule of the genetic plan.

#	Mechanism	State	Priority	Main action (suggestion, not applied)	Genetic findings
(mitochondrial redox cofactor)					
48	Redox imbalance	Mixed	Medium	Crucifers, glycine + NAC	NRF2 –617C/C, NQO1 *1/*1, GCLC and SOD2 favourable; Probable null GSTM1, GSTA1*B/*B, low expression PON1, MPO –463G/G and GCLM heterozygote.
49	Oxidative stress	Mixed	Medium	Crucifers, glycine + NAC; zero tobacco	The same pattern: strong antioxidant regulation, weak II phase conjugation.
50	Peroxisomal dysfunction	No reading	—	No useful genetic reading	No genetic reading.
51	Failure in Peroxisome-Mitochondria Communication	No reading	—	No useful genetic reading	No genetic reading.
Metabolism and endocrine					
52	Metabolic Homeostasis Decline	Mixed	Medium	Strength + aerobic; fiber	HbA1c genetics in the 7th percentile and fasting glucose in 9.5; 2 polygenic diabetes in the 57–69, TCF7L2 rs7903146 C/T, SLC30A8 and KCNJ11 percentiles in risk homozygosity.
53	Insulin resistance	Mixed	Medium	Low visceral fat; current metabolic stack	Fasting insulin in the 5th percentile (12 variants) and gluteal-femoral fat in 95; against trunk fat in 98, FTO, KLF14 and IRS1 heterozygotes.
54	Hyperglycemia and Intolerantness to Glucose	Favourable	Keep	Strength to maintain	HbA1c and genetically low fasting glucose; MTNR1B C/G (late meals are worse).
55	Dysfunction of adipose tissue	Mixed	Medium	Fat of the trunk p98,4: reducing visceral fat	Boundary fat in the 98th percentile and abdominal subcutaneous fat in 87; against gluteal-femoral fat in 95, adiponectin (loci) in 89 and FABP4 in 18.
56	Thermogenesis decline	Typical	Monitor	No specific action	UCP1 Ala64, ADRB3 Trp64 and DIO2 Thr92 favourable; UCP1 –3826 heterozygous and TBX15-WARS2 T/T.
57	Endocrine dysregulation	Favourable	Keep	Strength to maintain	Genetic testosterone in the 89th percentile, prolactin in the 12, typical thyroid High SHBG (the 87th percentile).
58	Decline of the Estrogen Signal	Typical	Monitor	No specific action	Little relevant in a man; ESR1 with common alleles
59	Thyroid alterations	Typical	Monitor	No specific action	Hypothyroidism and hyperthyroidism in the 34 and 38 percentiles; Low genetic TSH and high free T4 (without judgment).
60	Accumulation of Advanced Glycation Final Products (AGEs)	Favourable	Keep	Strength to maintain	HbA1c genetics in the 7th percentile: less baseline glycation.
Immunity, barriers, matrix and vessels					
61	Inflammaging	Mixed	Medium	The same; vaccines	CRP low and IL6R Ala (favourable) against GlycA in the 90th percentile and IL6 –174 G/G.
62	Immunosenescence	Mixed	Medium	Vaccines (pneumococcal, flu, zone); care with rapamycin	Good innate defense (MBL2 A/O without deficiency, TLR4 Asp299Gly absent, COVID-19 severe in percentile 4.7, sepsis in 3–23, neutrophils in 20); against polygenic tendency for bacterial mucosal infections (pneumonia in 84, urinary infection in 94, otitis in 85; zone in 75) and FCGR2A H131R heterozygous.
63	Thymic Involution	No reading	—	No useful genetic reading	No direct genetic reading (without score or variants with known effect on thymus).

#	Mechanism	State	Priority	Main action (suggestion, not applied)	Genetic findings
64	Loss of T Naive Cells	No reading	—	No useful genetic reading	Without direct genetic reading; the Pneumococcus and Immunosenescence report only conditionally binds to the weaker response to vaccines with age.
65	Repertory TCR restriction	No reading	—	No useful genetic reading	Without genetic reading (the repertoire is formed throughout life); HLA of class II typed only by markers.
66	Intestinal Barreira dysfunction	Favourable	Keep	Strength to maintain	Very low inflammatory bowel and celiac disease, secretory FUT2; CARD9 T/T and ATG16L1 T300A heterozygous.
67	Remodelling of the extracellular matrix	Mixed	Medium	Echocardiogram (aortic valve)	Discal disc and hernia very low (3–17 percentiles), COL3A1 and GDF5 favourable; COL5A1 rs12722 T/T, MMP9 and LOXL1 with weak risk alleles.
68	Stiffness of the Extracellular Matrix	Favourable	Keep	Strength to maintain	Arterial stiffness score (pulse wave velocity) in the 12th percentile.
69	Fragmentation of Elastin	Mixed	Medium	Avoid fluoroquinolones	Genetically wider ascending aorta (the 79th percentile; FBN1 and ULK4 with larger diameter alleles ELN and LOXL1 mixed).
70	Vascular Aging	Mixed	High	Low ApoB, aerobic fiber; coronary calcium between 35 and 40	Polygenic strengths (coronary disease in the 5th percentile, atrial fibrillation no 2, stroke no 14-19, dilated cardiomyopathy no 2) against focal weaknesses (9p21 homozygous, white matter lesions no 92, venous thrombosis no 86, aortic stenosis no 95, aorta no 79, venous thromboembolism no 95.1).
71	Arterial stiffness	Favourable	Keep	Strength to maintain	Arterial stiffness score in the 12th percentile.
Rhythms, sleep and brain					
72	Circadian disorders	Mixed	Medium	Light of the morning, fixed hours, morning chronotype	Morning Chronotype (the 86th percentile) and daytime sleepiness in 1 MTNR1B C/G and CLOCK G/A.
73	Desynchronization of Biological Clocks	Typical	Monitor	No specific action	No clear signal beyond MTNR1B (late meals).
74	Sleep disturbance	Unfavourable	High	Fixed time, TCC for insomnia, melatonin 0.3–1 mg	Insomnia (129 loci) in the 89th percentile, duration of genomic sleep in 17 and restless legs in 77.
75	Glymphatic dysfunction	Unfavourable	High	Regular deep sleep; healthy arteries	Perivascular spaces: SLC13A3 C/C (2 risk alleles, the strongest locus) and LPAR1 G/A; white matter lesions in the 92nd percentile.
76	Neuroinflammation	Mixed	Medium	Sleep, exercise, Omega 3	APOE ε3/ε3 and TREM2 without R47H/R62H; against microglial Alzheimer's loci (MS4A6A, EPHA1, PTK2B; 39 variant score in the 91st percentile) and central auditory pathway with a tendency to tinnitus (89.7 score, 94.6 gravity, 99.8 locus).
77	Hematoencephalic Barreira dysfunction	Unfavourable	High	Vascular protection; sleep; cognitive evaluation at 45–50	Small cerebral vessels: white matter in the 92nd percentile, PMF1 C/C and COL4A2 A/C; APOE ε3/ε3 attenuates.
Muscle, bone and frailty					
78	Sarcopenia	Unfavourable	High	Strength 3x/week + protein ~140 g/day; DEXA	Appendicular lean mass in the 10th percentile and fat-free mass in the 27–30 percentiles; sarcopenia "more likely" in an external report.
79	Thin Mass Loss	Unfavourable	High	Equal; protecting lean mass during retatrutide	The same sign of low lean mass; low genetic physical activity (7–14 percentiles).
80	Loss of muscle strength	Mixed	High	Progressive strength training; creatine	Typical genomic grip strength (51–52 percentiles), but load of force loci in 2.4.
81	Muscular atrophy	Unfavourable	High		

#	Mechanism	State	Priority	Main action (suggestion, not applied)	Genetic findings
				Strength + protein; avoiding immobilization	Low genetic lean mass; MSTN and ACTN3 typical.
82	Neuromuscular Junction Disorder	No reading	—	No useful genetic reading	No genetic reading.
83	Osteoporosis	Mixed	Medium	Impact, vitamin D and K2, DEXA	Fractures in the 84, ESR1, RSPO3 and SMAD3 percentile of risk; against ZBTB40, AXIN1, WLS and COL1A1 favourable and high genetic vitamin D (84–91 percentiles).
84	Fragility	Unfavourable	High	Strength, balance from 50, functional tests	Fragility in the percentiles 75 (943 thousand variants) and 82 (external report); cognitive fragility in 77.

3. Critical pharmacogenetics card

Base: 632 drugs evaluated with the guidelines CPIC and DPWG and with the regulatory labels. Result: 2 to avoid, 16 with dose adjustment, 35 to monitor or confirm; the others have a standard dose. Genotypes read by the chip and reconfirmed for this document, unless otherwise indicated. All the alternatives, adjustments and warnings of this section are suggestion (not applied) for the prescriber.

To show in any consultation, surgery or urgency

CYP2C9 *1/*3 (intermediate metabolizer) · VKORC1 –1639 T/T (Imputed, DR2 0.99; 1173C>T read in the chip is A/A, concordant; high sensitivity to coumarins) · NAT2 slow acetylator · UGT1A1 *1/*28 (Gilbert heterozygote) · CYP3A5 *1/*3 (expressor) · CYP2C19 *1/*1 (normal) · Probable normal CYP2D6 (no copy number) · Probable SLCO1B1 *1/*14 (without 521C of myopathy; increased function) · DPYD, TPMT, NUDT15 and G6PD normal · Normal BCHE · RYR1/CACNA1S without variants in the tested positions · SERPINA1 Pi*MZ (liver and lung) · genetic risk of high venous thrombosis · HLA-B*58:01 by typing (non-probable carrier)

3.1 Avoid (2)

Drug	Gene	Why	Alternative (sugesting)
Piroxicam	CYP2C9 *1/*3	Long-term half-life; exposure increases greatly (CPIC 2020: avoid with AS 1.0)	Short half-life NSAID at the minimum dose, or non-NSAID
Tenoxicam	CYP2C9 *1/*3	The same as piroxicam	The same

3.2 Adjust the dose (16)

Drug	Gene(s)	Adjustment (suggestion for the prescriber)
Warfarin	CYP2C9 *3 + VKORC1 T/T	Very high sensitivity: dose by pharmacogenetic algorithm (typically ~0.5–2 mg/day); prefer a DOAC when indicated
Acenocumarol, fenprocumon	CYP2C9 + VKORC1	~50% of the usual initial dose (DPWG)
Meloxicam	CYP2C9	~50% of the initial or alternative dose
Ibuprofen, celecoxib, flurbiprofen, lornoxicam	CYP2C9	Lower recommended dose, short course, avoid chronic use
Fluvastatin	CYP2C9	≤ 40 mg/day (current pitavastatin is not affected)
Phenytoin, fosphenytoin	CYP2C9	–25% in maintenance, with levels
Siponimod	CYP2C9 *1/*3	Maintenance 1 mg/day
Hydralazine, isoniazid	Slow NAT2	Lower doses; isoniazid has more hepatotoxicity, the worst combination with liver Pi*MZ + PNPLA3
Amifampridin	Slow NAT2	Lower initial dose
Tacrolimus (systemic)	CYP3A5 expressor	Initial dose 1.5–2x, with levels

3.3 Monitoring or confirming (choice of the most relevant)

Drug	Gene(s)	Note
Paracetamol	UGT1A1 *28; liver (Pi*MZ, PNPLA3)	Minimum effective dose, without alcohol, without prolonged fasting; avoid repeated maximum doses
Rapamycin (sirolimus)	CYP3A5 expressor; CYP3A4 normal	Lower exposure per mg; suggested dosage to calibrate. Avoid CYP3A4 inhibitors (clarithromycin, azoles, grapefruit)
Tramadol, codeine, tamoxifen, ondansetron	Probable normal CYP2D6	Standard dose, but the chip does not see duplications or deletions of CYP2D6; confirm in the complete genome
Irinotecan	UGT1A1 *28 heterozygous	A little higher toxicity; standard dose with surveillance
Sulfamethoxazole, dapsone, sulfasalazine, procainamide	Slow NAT2	More adverse reactions; monitoring
Methadone	CYP2B6 *2/*4 probable (fast); OPRM1 118 A/G	It may require a higher dose; monitoring the response

Drug	Gene(s)	Note
Isotretinoin, methotrexate, leflunomide, acitretin	Liver (Pi*MZ, PNPLA3)	Closer hepatic surveillance
Sulfonylureas	CYP2C9 *3	More hypoglycemia; low initial dose
Lithium (even in low dose) + NSAIDs	CYP2C9 *3	NSAIDs increase lithium in blood and CYP2C9*3 increases exposure to NSAIDs: avoid the combination
Allopurinol	HLA-B*58:01	Not probable by the imputed tag; type before starting (severe cutaneous reaction)
Carbamazepine, oxcarbazepine, phenytoin	HLA-A*31:01, HLA-B*15:02	A*31:01 probable by tag; B*15:02 not evaluated by the chip (very low a priori probability); type before possible
Abacavir	HLA-B*57:01	Negative; no restriction

Standard dose (normal genotype) in the following drugs: Clopidogrel, ticagrelor, prasugrel, proton pump inhibitors, citalopram/scitalopram, sertraline, most statins, ondansetron, fluorouracil and capecitabine (normal DPYD), azathioprine and mercaptopurin (normal TPMT and NUDT15), rasburicase and primaquine (normal G6PD).

3.4 Alerts in surgery, anesthesia and urgency

Situation	Alert (suggestion for the team)	Genetic basis	Evidence
General anesthesia	Succinylcholine and mivacurium with normal duration	Normal BCHE (rs1799807 T/T, rs1803274 C/C)	Strong
Malignant hyperthermia	Reduced but not absent risk: the chip tests part of the variants RYR1 (34 SNPs) and CACNA1S; standard precautions	No variants in the tested positions	Moderate
Postoperative	Prophylaxis of thrombosis to discuss always (early mobilization, heparin according to risk)	VTE p86, loci p95,1,F11 T/T	Moderate
Analgesia	Avoid piroxicam and tenoxicam; NSAIDs at the minimum dose and for a short time; paracetamol at the minimum effective dose (maximum 2 g/day)	CYP2C9 *1/*3; liver	Strong
Anticoagulation	If there is warfarin, very low dose per algorithm; prefer DOAC	CYP2C9 + VKORC1	Strong
Antibiotics	Avoid fluoroquinolones if there is an alternative (aorta); sulfamides with surveillance (NAT2)	P79 ascending aorta; NAT2	Moderate
Current medicine	Suspending dapagliflozin 3 days before surgery (euglycemic ketoacidosis); rapamycin delays healing (suspend around the surgery, deciding with the doctor); retatrutide delays gastric emptying (risk of aspiration; informing the anesthesiologist); tadalafil is incompatible with nitrates	Drugs (non-genes) next to the card because they change the conduct	Strong (labels and guidelines)
Mild jaundice in fasting or disease	It is probably Gilbert (benign)	UGT1A1 *28 heterozygous	Strong
Transfusion	The probable blood group (always confirm by typing)	ABO rs505922 T/T	Moderate

4. Current genes × stack (suggestions)

Only the genetic reading. The current dose of some items was decided with clinical data that this document does not use; when genetics points out another direction, this is said, but the clinical decision prevails. Everything is suggestion (not applied).

4.1 Medicines

Item	Relevant genes	Genetic reading	Suggestion (not applied)
Pitavastatin 4 mg, ezetimiba, bempedoic acid	SLCO1B1 rs4149056 T/T (without 521C; Probable *1/*14, increased function; ABCG2 G/G; 9p21 homozygous	Without the SLCO1B1 myopathy allele, the risk of myopathy does not increase; normal ABCG2 does not predispose to high urate with the wellpedoic	Keep ; is the n.o 1 defense against 9p21. Discuss to resume the inclisiran or other anti-PCSK9 and, in the future, the obicetrapib (high activity CETP)
Rapamycin 4 mg of 6 in 6 weeks	CYP3A5 *1/*3 expresser; CYP3A4 normal	Less exposure per mg; genetic pneumonia p84,2	Dose to calibrate; pneumococcal conjugated before; suspension in infections and around surgeries; keep the physetin ≥ 7 days away
Isotretinoin 40 mg/week	Pi*MZ, PNPLA3, MBOAT7; p95 depression	Vulnerable liver; more risk of mood	Regular liver surveillance; monitoring mood; photoprotection; no vitamin A or retinol
Retatrutide 8 mg of 6 in 6 days	P10 lean mass; p75 fragility; p98,4 trunk fat	Useful for visceral fat, but lean mass is the weak point	Protein ~140 g/day + strength 3×/week; DEXA to follow lean mass
Dapagliflozin	Urinary infection p93, 8; normal UGT1A9	More tendency for urinary and genital infections	Hydration and hygiene; suspend 3 days before surgery; care with ketogenic diets
Imeglimine, acarbose	TCF7L2 C/T; diabetes p57	Coherent with moderate metabolic risk	Keep
Tadalafil 5 mg	Normal CYP3A4; high genetic tinnitus	No adjustment; attention to sudden hearing loss (rare)	Maintain; never with nitrates
Naltrexone 4.5 mg	OPRM1 118 A/G	No adjustment established	Keep
Telmisartan 20 mg	No actionable gene	No relevant genetic interaction	Maintain; potassium only by food (without KCl tablets)
Lithium 5 mg	CYP2C9 *3 (NSAIDs)	NSAIDs increase lithium	Avoid NSAIDs; hydration

4.2 Supplements

Item	Genes	Genetic reading	Suggestion (not applied)
Fish oil	FADS1 rs174546 C/C (efficient conversion); 9p21; liver	Cardiovascular and hepatic benefits	Up to 3.2–4 g of EPA+DHA/day (already taking fish oil; the product of the plan has EPA 1125 + DHA 875 mg), form rTG, with meal
5-MTHF 800 µg + B12 1000 µg + B2	MTHFR C677T G/A + A1298C T/G; genetic homocysteine p92	Slower methylation pathway	Maintain; never folic acid; B2 can decrease to 5–10 mg
B6 15 mg	PDXK G/G; typical need	Above the upper limit of long-term security	Drop to ≤ 5 mg or stop (peripheral nerve injury with prolonged use)
TMG 1,5 g	BHMT G/A; GNMT T/T	Useful betaine pathway	Up to 3 g/day in 2 tomas (maximum 3 g, not to rise LDL)
Glycine 3 g	CPS1; Probable null GSTM1; GSTA1 *B/*B	Glutathione and sleep	Up to 5 g/day (3 g at bedtime + 2 g in the morning)
NAC 1800 mg	GSTM1/GSTA1	Glutathione support in the weak II phase	Keep
Vitamin D3 10 000 IU + K2 240 µg	GC (genetic vitamin D p84,7); VKORC1 T/T	The isolated reading would arrive with ~4000 IU	Keep the clinical decision; K2 compatible (without warfarin)
Zinc and copper	Zinc and typical genetic copper	—	Total zinc 20–25 mg/day with copper 1.5–2 mg (proportion ~10–15:1)

Item	Genes	Genetic reading	Suggestion (not applied)
Creatine 5 g, collagen 20 g, taurine 6 g	Thin mass p10; matrix	Muscle and connective tissue support	Keep
Psyllium 10 g + PHGG 6 g	Colorectal p94,3; diabetes	Fiber: more is always better	Up to psyllium 15-20 g + PHGG 10 g , gradually
Lycopene 35 mg	Low genetic prostate	No genetic reason for high dose	To go to 10 mg or stop
Melatonin (up to 100 mg)	Insomnia p89; circadian clock	The high dose loses the circadian specificity	0,3-1 mg , 30-60 min before bedtime and ≥ 3 h after the last meal (MTNR1B C/G)
Lutein and zeaxanthin	CFH Y402H C/T	Macula	Keep
Sulforaphane, aged garlic, ergothione	Probable null GSTM1; slow NAT2	Induction of II phase	Keep
Vitamin C 500 mg	HFE H63D C/G	Increases iron absorption	Keeping away from meals
Curcumin, luteolin, apigenin, quercetin	Liver (Pi*MZ, PNPLA3)	Rare reports of liver injury with concentrated extracts	Suggestion (not applied): pause
Epitalon	TERT rs2736100 C/C (glioma p91)	Peptide presented as telomerase activator	Suggestion (not applied): pause
Peptide remains (SS-31, MOTS-c, humanin, thymosine α 1, thymaline, FOXO4-DRI, oxytocin, pinealon)	No actionable gene	Weak human evidence in all	No specific genetic objection; clinical decision

4.3 Avoid for genetic reasons

Avoid	Why (genes)
Folic acid	MTHFR compound: prefer 5-MTHF
Iron and multivitamins with iron	HFE H63D; vulnerable liver
Preformed vitamin A and retinol	Isotretinoin + liver
Berberine	CYP3A4/2D6 interactions and rapamycin
BCAA isolates	No benefit beyond the complete protein; mTOR signal
Secretagogues of GH/IGF-1, telomerase activators	TERT C/C (glioma p91); cancer
Melanotan, BPC-157, TB-500	No safety data; melanotan with skin risk
Grapefruit	It inhibits CYP3A4 (rapamycin, tadalafil)
Herbal supplements with hepatic risk	Pi*MZ + PNPLA3
Potassium in tablets	With telmisartan

5. Screening calendar

Three phases: Now, next years and later. The ages that appear are the generic target ages of the screening recommendations. The general screenings of the population are included; those that were anticipated or added by the genes are the bold. The intervals are the usual starting point: an abnormal finding shortens them. Everything is suggestion (not applied).

Now, once

Examination	Repetition	Why	Evidence
Serum Alfa-1-antitripsina + Pi phenotype	Once	Confirm Pi*MZ (chip)	Strong
Transient hepatic elastography	2 in 2 or 3 in 3 years	Pi*MZ + PNPLA3 + MBOAT7; isotretinoin	Strong
Echocardiogram (aortic valve and left ventricular wall thickness)	40, after 5 in 5 years	Aortic stenosis p89-95; hypertrophic cardiomyopathy p76-89	Moderate
Spirometry	40, after 5 in 5 years	P96,2 + Pi*MZ	Moderate
Ophthalmological examination with intraocular pressure + OCT of the optic nerve	From 2 in 2 years to 40; annual after	Glaucoma p97, 9	Moderate
Audiogram	5 in 5 years; annual from 50	Tinnitus (loci) p99, 8	Weak to moderate
Whole body DEXA (lean mass + bone density)	2 in 2 years or when changing the medication	P10 lean mass; p84 fractures	Moderate
Complete genome to clinical 30 (ACMG, pharmacogenetic with CYP2D6 and HLA)	Once	Closes the chip gaps (8 section)	Strong
HLA-B*58:01 typing (if not found in the genome)	Once	Before any allopurinol	Strong
Lp(a)	Once in life	Strong alleles are absent, but KIV-2 is not read	Strong (general recommendation)
Dermatology	Annual while doing isotretinoin; after 2 in 2 years	Low genetic melanoma; drug photosensitivity	Moderate
Dental and oral hygiene	6 in 6 months	Gingivitis p89, 7	Moderate
Mood and sleep questionnaires (PHQ-9, GAD-7, ISI)	6 in 6 months	P95 Depression; p89 insomnia	Moderate
Vaccines (section 6.5)	—	Pneumonia p84, 2; rapamycin; liver	Strong

Next years

Examination	Repetition	Why	Evidence
Plan revision with the complete genome results	Once	Integrate new findings	—
Elastography, DEXA, ophthalmology	According to the above intervals	—	—
Evaluation of ADHD, if never done	Once	PHDA p94/p97	—
Coronary calcium score (TC without contrast) between 35 and 40	If 0: 5 in 5 years; if > 0: angio-TC and intensify	9p21 homozygous	Moderate to strong
Review of the targets of ApoB and LDL	Annual	Lower is always better	Strong
FIT (immunochemical occult blood) from 40	Annual (or 2 in 2 years) until colonoscopy and at intervals	Colorectal p94,3; early onset p82,3	Moderate
Echocardiogram from 40	5 in 5 years	Aortic valve; hypertrophic cardiomyopathy	Moderate
Spirometry from 40	5 in 5 years	Pi*MZ	Moderate

Examination	Repetition	Why	Evidence
Ophthalmology with OCT	Annual from 40	Glaucoma p97, 9	Moderate
Urine type II (hematuria) from 40	Annual	Slow NAT2 + bladder p90	Weak
Panel of clonal hematopoiesis (CHIP) in blood between 40 and 50	5 in 5 years if negative	Acquired; the chip does not see it	Moderate
Colonoscopy of quality (maximum to 45)	The findings (typically 5–10 years; 3 if there are advanced adenomas)	Colorectal p94,3, genomic p78,5, polyps p92,7	Moderate
Cognitive assessment between 45 and 50	5 in 5 years	White substance p92	Weak to moderate

Later

Examination	Repetition	Why	Evidence
PSA by shared decision from 50	2 in 2 years if you decide to do	Low prostate genetic risk	Moderate
Audiogram from 50	Annual	Tinnitus and hearing loss	Weak to moderate
Functional tests (handgrip strength, chair lift, balance) from 50	Annual	P75 fragility	Moderate
Ankle-arm index from 50	To be discussed; 5 in 5 years	9p21 (peripheral arterial disease)	Weak
Recombinant zone (if not yet made)	2 doses	Routine from 50	Strong
Colonoscopy or FIT	As the last result	Colorectal	—
Densitometry (DEXA)	2 in 2 years	P84 fractures	—
Echocardiogram between 55 and 65	From 3 in 3 to 5 in 5	Aortic stenosis p89–95	—
RSV from 60	According to the recommendations	Pneumonia p84, 2	—
Pneumococcal (review the scheme) from 65	According to the recommendations	Pneumonia p84, 2	—
Echography of the abdominal aorta to 65 (general screening recommended for men)	Once	General population screening	—
Evaluation of falls and fragility from 65	Annual	P10 lean mass; p84 fractures	—
Cognitive evaluation from 65	2 in 2 years	White substance p92	—
Colorectal tracking to 75 (individualization later)	According to the findings	Colorectal	—

6. Food, exercise, sleep, environment and vaccines

6.1 Food

Principle	Genes	Evidence	Suggestion (not applied)
Fiber: more is always better (target ≥ 40 -50 g/day with supplements)	Colorectal p94,3; polyps p92,7; diabetes p57; 9p21	Strong (fiber) / moderate (genes)	Daily legumes, oats and barley, vegetables, whole fruit, dried fruits, seeds; psyllium and PHGG rising
Low saturated fat; olive oil, fatty fish and dry fruits as main fat	9p21; CETP of high activity; HDL p17-20	Strong	Fatfish 2-3×/week; no butter or lard as base
Little processed meat and little red meat; no carbonized meat	Colorectal; Slow NAT2 (heterocyclic amines)	Moderate	Steam cooking, ovening or roasting at low temperature
Daily crucifers (broccoli, kale, rocket, cress)	Probable null GSTM1; GSTA1 *B/*B; Slow NAT2	Moderate	Natural induction of II phase
~140 protein g/day in 3-4 meals of 30-40 g	P10 lean mass; p75 fragility	Moderate	Fish, eggs, dairy products (persistent lactase: LCT A/G), legumes, soybean
Zero alcohol (already fulfills)	Pi*MM + PNPLA3; drinks per week p94,6	Strong	Keep
Coffee filtered up to ~200 mg of caffeine/day, only in the morning	Insomnia p89; CYP1A2 *1F A/C (normal elimination) and ADORA2A C/C (less sensitive): the limit comes from sleep and plane	Moderate	None after 12-13 h
Iron only from food; vitamin C far from iron-rich meals	HFE H63D C/G; liver	Moderate	No iron supplements
Food folate (leaf vegetation)	MTHFR compound	Moderate	5-MTHF instead of folic acid
Grapefruit: avoid	CYP3A4 (rapamycin, tadalafil)	Strong	—
Gluten	HLA-DQ2.5 absent; celiac p4	Moderate	No genetic reason to avoid it

6.2 Exercise

Component	Genes	Suggestion (not applied)
Strength 3×/week, all major muscle groups, with progression	P10 lean mass; p84 fractures; ACTN3 T/C	It is the 1 intervention for the weakest point of the body
2 zone aerobics 150-300 min/week + 1 interval session	Aerobic resistance p92; 9p21; white matter p92	Taking advantage of aerobic genetic strength
Impact and jumps (2-3/week, low volume)	P84 fractures	Bone
Balance and mobility from 50	P75 fragility	Prevent falls
Move 30 in 30-60 minutes	Genetic physical activity p14, 2; VTE p86	Genetics pushes to sedentary lifestyle: compensating with routine
Tendon care	Typical collagen variants	Gradual progression

6.3 Sleep and circadian rhythm

Principle	Genes	Suggestion (not applied)
7-9 hours with fixed time, also at the weekend	P89 insomnia; duration of sleep p17; morning chronotype p86	Lie down and wake up early, in favor of the genetic clock
Natural light in the morning (10-30 min); little light at night	Circadian clock; mood	No screens in the last hour
TCC for insomnia before any hypnotic	P89 insomnia; p95 depression	First line
Melatonin 0,3-1 mg	—	Physiological dose
Dark, fresh and silent room	White substance p92 (glymphatic)	—

Principle	Genes	Suggestion (not applied)
Search sleep apnea if there is strong snoring or drowsiness	Fat of the trunk p98, 4	Polysomnography if there are signs

6.4 Environment and exhibitions

Everything in this section is suggestion (not applied).

Avoid or reduce	Genes	Why
Smoking of any kind (tobacco, vaping, cannabis, chimney without ventilation)	9p21, Pi*MZ, emphysema p96, 2, bladder p90 + NAT2	The external factor that most adds to the genetic weaknesses
Amines and solvents (dark permanent hair dyes, rubber dyes, diesel)	Slow NAT2; probable null GSTM1	Bexiga
Powders and fumes without FFP2 mask	Pi*MZ (lung)	Emphysema
Noise (> 85 dB) without protection	Zombido p99, 8	Hearing and dementia
Excessive sun during isotretinoin	Drug	Photosensitivity (genetic melanoma is low)
Unneeded medical radiation	ATM D1853N A/A	Prefer RM or ultrasound when equivalent
Extreme heat with dehydration	Dapagliflozin, lithium	Hydration

6.5 Vaccines

Vaccine	When	Why	Evidence
Conjugated pneumococcal (PCV20 or PCV21)	Já (to be discussed), single dose	Intermittent immunosuppression (rapamycin); pneumonia p84, 2; FCGR2A H/R	Moderate
Influenza	Annual, in autumn	Pneumonia p84, 2	Strong
COVID-19	According to the recommendations in force	Rapamycin	Strong
Hepatitis B	If not immune	Vulnerable liver (Pi*MZ + PNPLA3)	Strong
Recombinant zoe	Discussing earlier (immunossupression); routine to 50	Rapamycin	Moderate
dTpa	10 in 10 years	Routine	Strong
HPV	Discussing (up to 45)	Prevention of associated cancers	Moderate
VSR	From 60	Pneumon	Moderate
Live vaccines (yellow fever, live varicella)	Avoid during rapamycin	Immunosuppression	Strong

7. Genetic forces to take advantage

Strength	Genes and Scores	Evidence · trust	How to take advantage (suggestion, not applied)
APOE ε3/ε3, TOMM40 and MAPT H2/H2	Without ε4 (the largest common genetic factor of mortality and Alzheimer's); TOMM40 rs2075650 A/A; MAPT H2/H2	Strong · imputation (DR2 0.99) + chip	The brain starts from a good base: the work is vascular (small vessels) and sleep
FOXO3 of longevity in homozygosity; low IGF-1 axis	FOXO3 with 4 SNPs of homozygosity (rs2802292 G/G, rs2764264 C/C and more 2); Low to intermediate IGF-1 (52 loci p7; genomic p34)	Strong (FOXO3) / moderate · chip	Do not "pull" the GH/IGF-1 axis with secretagogues; maintain insulin sensitivity
High polygenic longevity	PGS of longevity p92	Moderate · PGS	—
Low polygenic coronary risk	metaGRS p5; LDL polygenic p20,5; p2 atrial fibrillation; Stroke p19; without the strong alleles of Lp(a); factor V and prothrombin negative	Moderate to strong · PGS + chip	With lipid therapy, 9p21 is highly compensated
Low glucose and glycation	HbA1c genetic p6, 6 and p8, 4	Moderate · PGS	—
Almost all cancers with low risk	Prostate p0,9; kidney p0,8; lymphoma non-Hodgkin p1,1; squamous cell p3,5; melanoma p6,5	Moderate · PGS	Exception: colorectal (1.2 section)
Low autoimmunity and inflammation	IBD p2; celiac p4 (HLA-DQ2.5 absent); CRP genetic p12; COVID-19 severe p4,7	Moderate to strong · PGS + chip	No genetic restrictions
Rim	eGFR genetic p94	Moderate · PGS	—
Blood: few clonal expansions	CHIP p30 score; RDW genetic p3,9	Moderate · PGS	Still, the CHIP panel to 40-50 (is acquired)
Little vulnerability to tobacco	"Having smoked" p2	Moderate · PGS	—
Aerobic resistance	Aerobic resistance p92; genetic resting heart rate p4,4	Moderate · PGS	Use the aerobic as an anchor of vascular and cerebral health
Liver: partial protection	HSD17B13 rs72613567 T/TA; TM6SF2 C/C	Strong · imputation (DR2 1) + chip	Do not dispense the protection of the liver
Favourable pharmacogenetics in many ways	CYP2C19*1/*1, SLCO1B1 without myopathy allele, DPYD/TPMT/NUDT15/G6PD normal, BCHE normal	Strong · chip	Most common drugs in standard dose
Morning Chronotype	Morning Chronotype p86	Moderate · PGS	Early hours, in favor of the clock

8. What the chip does not see and how to close the gaps

Everything in this document comes from a genotyping chip (about 685 thousand positions) completed by imputation. There are four types of information left out that only the sequencing reads.

1. Genome complete to 30x (germinal)

What is missing	Why does it matter here
GSTM1 and GSTT1 deletions	The null GSTM1 is only "probable" and the GSTT1 is not testable; both count on detoxification, which is already the weak point of the II phase (slow NAT2, GSTA1*B/*B).
Rare variants of hereditary disease	Only 21% of the P/LP variants known in 81 ACMG genes are read (35% of those with ≥ 2 stars); the large deletions (BRCA1, MSH2/EPCAM, LDLR) and the new variants are not seen. A negative result does not exclude hereditary disease.
HLA	There are only imputed markers (B*58:01, B*57:01 and A*31:01 probable negative); the HLA typing of 2 fields gives certainty before allopurinol, abacavir or carbamazepine and completes the autoimmune profile.
Complete mitochondrial DNA	Only classical 4 variants and haplogroup were read; heteroplasma and rare variants require sequencing.
Structural variables and repetitions	Number of copies of CYP2D6, kringle repetitions IV-2 of LPA (the main determinant of Lp(a)), APOL1 G2, SMN1, alpha-thalassemia, polyT of TOMM40, repetition CAG of the androgen receptor and expansions (C9orf72, HTT, FMR1).
Variants of rare centenarians	centSIRT6 (N308K/A313S), ANGPTL8 Q121X and most protective function losses of APOC3, ANGPTL4 and PCSK9 are not read (from the 343 variants of ClinVar in these genes, only 19 were read, all absent).
BMPR2 and arterial pulmonary hypertension genes	Of the ~1970 pathogenic variants known in BMPR2 and in more 11 related genes, the chip only covers a small fraction: BMPR2 is not evaluated.

2. Clonal hematopoiesis (CHIP)

CHIP are clones of blood cells with acquired mutations (mainly DNMT3A, TET2, ASXL1, JAK2 V617F). It appears in 10–20% of people over the age of 70, almost doubles coronary disease and increases mortality and the risk of blood cancer. It is somatic: the chip reads the inherited predisposition (typical to low: without JAK2 46/1, SH2B3 C/C, CHIP score in the 30th percentile; TCL1A G/G and TERT pulling upwards), but not the clones. To see it, it is necessary to sequence the blood with a panel of high-depth CHIP mutations (detector clones from ~1–2% of cells). With 9p21 homozygous, a clone TET2 or DNMT3A would add vascular risk.

3. Other measurements that the inherited DNA does not replace

- Epigenetic clocks measured in blood (the real biological age, instead of the inherited loci).
- Length of the measured telomeres (the genetic score only gives the trend).
- Lp(a) measured once in life.
- Somatic mutations in other tissues and loss of Y in the blood.

4. How to close, in order

Everything is suggestion (not applied).

Step	What does	When
Complete genome to clinical grade 30 , with a report of secondary ACMG and pharmacogenetic findings (including the number of copies of CYP2D6 and HLA)	Rare hereditary disease, GSTM1/GSTT1, CYP2D6, HLA, KIV-2 of LPA (if the laboratory reports), BMPR2, DNA mitochondrial	Now
Phenotype Pi + serum alfa-1-antitripsina	Confirms Pi*MZ by another route	Now
HLA typing of 2 fields (if not found in the genome)	Certainness before allopurinol, carbamazepine and others	Now
Lp(a) measured once	Closes KIV-2	Now
CHIP panel in blood	Acquired clones (DNMT3A, TET2, ASXL1, JAK2)	40–50 years; 5 in 5 years
It is solved with the complete genome		

9. Plan by horizons

Everything is a suggestion (not applied), to decide with the doctor.

Already (next 2-4 weeks)

1. Keep the pharmacogenetic card (3 section) on the cell phone and deliver it to the family doctor.
2. To ask alfa-1-antitripsina + phenotype Pi and hepatic elastography.
3. Mark the conjugated pneumococcal and confirm the immunity to hepatitis B (vaccine if not immune).
4. Simple adjustments to the stack: B6 for ≤ 5 mg; melatonin for 0.3-1 mg; lycopene for 10 mg; glycine for 5 g; TMG for 3 g; fish oil for 3.2-4 g of EPA+DHA (already taking fish oil; climbing to this target).
5. Fix sleep time and morning light; coffee only until 12-13 h.
6. Force training plan 3x/week with load recording.

3 months

1. **Complete genome to clinical 30** (with ACMG, CYP2D6 and HLA), or at least the HLA typing.
2. **Echocardiogram, spirometry, OCT with intraocular pressure, audiogram, DEXA** (base package).
3. Fiber to gradually rise to psyllium 15-20 g + PHGG 10 g.
4. Mood and sleep questionnaires; TCC for insomnia, if there is insomnia.
5. Discuss with the doctor: resuming the inclisiran or other anti-PCSK9; measurement of rapamycin; hepatic and humoral surveillance with isotretinoin.
6. Lp(a) once.

1 ano

1. Review this document with the complete genome result (ACMG, CYP2D6, HLA, GSTM1/GSTT1, KIV-2, BMPR2).
2. Repeat DEXA if lean mass is the focus; adjust protein and training.
3. To review the vaccines (recombinant zone to be discussed by rapamycin; HPV to discuss).
4. Reviewing the stack of supplements with the doctor (items without genetic or clinical basis can leave).

Continuous (all life)

- **ApoB and LDL as low as possible**, without long breaks; zero smoking; zero alcohol.
- Strength + aerobic + protein; regular sleep; high fiber.
- Schedule of screenings of section 5 (coronal calcium between 35 and 40, FIT at 40, colonoscopy at 45 at the maximum ...).
- Prophylaxis of thrombosis in each surgery, immobilization or long journey.
- Show the pharmacogenetic card before any new drug.
- Auditory and ocular protection; 6 dentist in 6 months.
- Update the plan when there are new genetic data (complete genome, better scores).

10. New DNA analysis for this document

For this document, in addition to rereading all the reports, I returned to the raw DNA and to the imputation and made new 5 analyzes.

10.1 ClinVar screening of the entire genome (pathogenic and probably pathogenic variants)

- Base: ClinVar of 28-09-2026, without the "conflict" classifications; 355530 variants P/LP in total.
- Read: 24 398 (6.9%), of which 24 360 by chip and 452 by imputation ($DR2 \geq 0.8$; some by two routes). In the variants with ≥ 2 stars: 16 901 of 101 547 (16.6%).
- **Only plausible P/LP finding: SERPINA1 Pi*Z heterozygous** (already known; reconfirmed).
- Other signs, all explained:
- FLG rs542799026G/C possible (skin and eczema; already known);
- BRCA1 and KCNH2 indel probes, already explained by common neighboring variants;
- SBDS is a known artifact (gene with pseudogene).
- **6 calls of ultra-rare variants of severe childhood diseases in 2 copies** (5 homozygotes: GBA1, IDUA, ALG1, NF1, GOSR2 1 hemizygote in the X chromosome: OFD1), all absent from the reference panel: incompatible with a healthy adult, so they are probe artifacts. They are confirmed or excluded in the complete genome.
- The common variant of TG (rs2294024; frequency ~ 0.48) and the SNPs of 9p21 with low confidence classification are not monogenic.
- **Conclusion:** In what the chip and the imputation read, there is no severe monogenic disease. The coverage is partial: a negative does not exclude (8 section).

10.2 ClinVar variants classified as "protective"

- 53 variants evaluated; 30 read; carrier in 14.
- No major effect (such as PCSK9, APOC3 or ANGPTL3 function losses) was observed.
- Highlights, all with weak evidence: CASP8 -652 6N of heterozygous (cancer), ADH1C, FCGR2B, promoter of CCR5, TLR1, NOS3, BMAL2, CYP46A1, CD209.
- The "protective" labels of 3 SNPs of 9p21 contradict the link in the reference population: they are the same risk haplotype (see 10.4).

10.3 Reconfirmation of 66 critical genotypes

All the same as what was in the reports:

- 9p21 and SERPINA1;
- liver: PNPLA3, MBOAT7, TM6SF2;
- APOE and FOXO3;
- MTHFR, F5, F2 and F11;
- complete pharmacogenetic panel: CYP2C9, VKORC1, CYP4F2, CYP2C19, CYP3A5, CYP3A4, SLCO1B1, NAT2, CYP2D6, DPYD, TPMT, NUDT15, UGT1A1, BCHE;
- HFE, LPA, TCF7L2, ATM, CFH, TERT and TP53.

The **HSD17B13 rs72613567 T/TA** was confirmed directly in the imputation ($DR2$ 1, heterozygous genotype), in addition to the chip marker.

10.4 Connection (LD) in the 9p21, recalculated

- In 633 genomes of the reference population, the risk 5 SNPs (rs4977574, rs1556516, rs1333042, rs10757278, rs1333049) have correlations r from 0.92 to 0.998: they are a single haplotype.
- The titer is homozygous at risk in 5. This counts as a signal (2 copies of the risk haplotype), not as independent 5 signals.

10.5 Traceability table

Each numerical statement of this document is linked to a section of the genetic profile and to the corresponding rsID or score, for independent verification.

11. Contradictions between reports and the choice made

General rule: it is worth the data with more variants, calibrated in the reference population and read directly (chip > imputed > score of few variants > external report without published method).

#	Topic	What do the sources say	Choice and why
1	Colorectal cancer	External report p76 → small scores p48,7 and p55,6 → complete scores p94,3 and p78,5, early onset p82,3, polyps p92,7	Values of complete scores (more variants, calibrated). The old reading "not justify anticipation" was replaced: colonoscopy to the 45 at most + FIT from the 40
2	Melanoma	External report p89 vs chip p6, 5 and p25, MC1R clean	It is worth the chip. Photoprotection is maintained because of isotretinoin
3	Prostate	External report p79 vs genomic p0,9 without the strong loci of 8q24	It is worth the chip. PSA only by shared decision from 50
4	Telomeres	"shorter" external report → p49 → p24 (7 SNPs) → p25 (87 loci)	Slightly short; weak evidence; no action beyond the lifestyle (and nothing of telomerase activators)
5	Alzheimer	External report p40 / 39 variants p91 / genomic p52, with APOE ε3/ε3	Genomics (with APOE, the major determinant). The real point is the white matter p92 (vascular)
6	Lp(a)	PGS p75 vs the 2 strong alleles (rs10455872, rs3798220) absent; KIV-2 not read	Lp(a) genetics unlikely; measure once in life
7	Coronary	Low polygenic (metaGRS p5, GPS p30) vs 9p21 homozygous	Both are true: low global basis with a strong focal point. The action follows the 9p21 (ApoB low all life)
8	2 type diabetes	71 → 57 after imputation	57 Valley (more variants read)
9	Liver fat	Fatty liver scores between p23, 9 and p78, 4; liver fat (resonance) p65, 4 and p69, 4 — all typical, despite PNPLA3 and Pi*Z	Large-effect alleles (PNPLA3, MBOAT7, Pi*Z) define the risk of fibrosis; HSD17B13 protects. The action follows the big alleles
10	Longevity	PGS p89 → p92	92 Valley (Imputation version)
11	rs12785878 (DHCR7)	External report G/G vs chip T/T	Validity of the chip
12	TP53 codon 72	Old label "Pro72Pro" vs Arg72/Arg72 for the same genotype C/C (Imputed, DR2 0.97)	It was a label error: in strand + C encodes Arg72. Arg72/Arg72 valley (more apoptosis, less longevity in some studies; compromise)
13	HLA-B*58:01	Indeterminate → non-probable carrier (imputed tag)	Non-probable carrier, but typing before allopurinol
14	9p21 "protector" in ClinVar	rs1333042 and others with "protective" vs LD r 0.92–0.998 label with risk SNPs	It is worth the connection recalculated: is the haplotype of risk
15	Fragility	p82 vs p75	Concordant; p75 is used
16	Pneumon	p38 (immunity report) vs p84,2 (most recent genomic score)	Values the genomic score p84,2 (more variants); pneumococcal vaccine is suggested in any case (rapamycin)
17	Melanoma and melanotan	The "avoid melanotan" was based on the external p89	The genetic reason weakened, but it keeps avoiding (without security data)

12. Limitations and methods

Methods

- **Sources read:** the complete genetic profile (106 sections), the favourable/disfavorable summary, 8 external reports, the 86 thematic reports, the genetic plan (103 items), pharmacogenomic (632 drugs), supplements (999) and foods (812).
- **Genetic data:** Genotyping chip (about 685 thousand positions) and imputation for the complete genome, against a reference panel of 30× genomes ($DR2 \geq 0.8$ quality to use an imputed variant).
- **Polygenic scores:** PGS Catalog, plus scores constructed from the GWAS Catalog; percentiles calculated with exactly the same variants in 633 genomes of the reference population.
- **New analysis (10 section):** ClinVar of the entire genome, "protective" variants, 66 critical genotypes, HSD17B13 in imputation and binding of 9p21.
- **Ordering:** Frequency of the cause in the population × genetic deviation × strength of evidence × possibility of acting.
- **Evidence:**
 - **strong:** large effect variant, or CPIC/DPWG guideline, or locus replicated with mechanism;
 - **moderate:** PGS well validated, or replicated locus without large effect;
 - **weak:** a single study, small effect or little validated score.
- **Trust:**
 - **chip:** genotype read directly;
 - **imputed:** estimated, with $DR2 \geq 0.8$ quality;
 - **PGS:** statistical trend, non-diagnosis.

Limitations

1. **The chip is not a genome.** It reads 6.9% of known P/LP variants and 21% in the ACMG genes. A negative does not exclude a hereditary disease.
2. **PGS explain only one part of the risk**, especially in psychiatric diseases and cancer. A high percentile increases the probability, but does not say that the disease will happen, nor when.
3. **The percentiles are related to the reference population.** They are not absolute risk. To have absolute risk it would be necessary to combine age, sex and clinical factors, and this document, on purpose, does not use them.
4. **Structural variants, repetitions and somatic mutations are not seen:** CYP2D6 with copy number, KIV-2 of LPA, GSTM1/GSTT1, CHIP.
5. **Gene-gene and gene-environment interactions** are poorly studied. The sum of points (e.g. $Pi^*MZ + PNPLA3$) follows the literature, but there is no validated model to combine them.
6. **Only genes.** This document does not use blood tests or exams. Current clinical decisions may have reasons that do not appear here and prevail over genetic reading.
7. **Nothing here is a prescription.** These are suggestions (not applied) to discuss with the doctor.